

Session 9 • Thursday 24 September 2026 • Tuesday's ledger stays up. Two lines join it. The group answers go on the right wing and are named back to the room.

Board notes

Print this and carry it. The deck records what is *projected*; this records what gets *written*.

This session is not a board session. One derivation, on the split surface. The board carries **Tuesday's ledger**, rewritten before class if it was wiped, plus **two new lines**, plus the **group answers** from the argument at 16–26.

Conventions. **ASK** — put it to the room before writing, and the answer to expect. **POINT** — the moments you turn and point at the board rather than the screen. **IF ASKED** — the question that always comes.

Before class — left wing, Tuesday's ledger, rewritten

$$\frac{du}{dt} = \frac{\alpha_1}{1 + v^\beta} - u \quad \frac{dv}{dt} = \frac{\alpha_2}{1 + u^\gamma} - v$$

fixed points = crossings

saddle $\iff g_1 g_2 > 1$

Written in the paper's notation today — $\alpha_1, \alpha_2, \beta, \gamma$ — because the numbers on the board are pTAK117's: **156, 15.6, 2.5, 1**. Write those under the equations.

Right wing — minutes 16–26, the group answers

Three columns, one per question, answers **by group name**. You read them back at 26 and sort them into the four rows of the next surface. Do not sort while writing.

ASK at 26, before sorting: *which of these did every group have?* Mutual repression. *Which did one group have?* Removal — or nobody, which is the usual case, and then it is the amber row.

Left wing, line 1 — minute 45, the last step of the run

$$s_c = 5.4 \iff 39 \mu\text{M IPTG} \quad (K = 2.96 \times 10^{-5} \text{ M}, \eta = 2.0)$$

POINT at the 15.6 and the 156 as you write it. Nothing about the circuit was changed to get this number; it is the legend's own parameters and a count of crossings. Then turn to the screen for Fig. 5a and point at the jump.

ASK: *what did the IPTG do to the picture?* One curve stretched, the other did not move. If the room says "both moved", go back one step.

IF ASKED *why does it not come back when IPTG is removed?* — because the high state exists at zero IPTG; it always did. Nothing pulls the cell off a state that exists. That is why the paper needed a second inducer.

Left wing, line 2 — minute 60, after the ConcepTest

threshold $\leftarrow \alpha, \beta, \gamma$

switching time $\leftarrow$ removal

ASK before writing the right half: *what was measured in Fig. 6 — a threshold or a time?* A time. Three to four hours to begin and done by six going up; thirty-five minutes down — and the difference is dilution against heat destruction.

IF ASKED *does a tag on one repressor change the threshold too?* — yes. In scaled units removal sits inside α , and the Jacobian is no longer symmetric. That is item 4 of Tuesday's handout and PS4 Q5.

What does not go on the board

- The run. It is on the split surface with the picture beside it.
 - The four failure mechanisms before the room has argued. Foot line, after.
 - Any fixed-point arithmetic from the faded set. Circulate instead.
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After class

Photograph the left wing. Tuesday's four lines plus today's two are the week, and the first thing to check PS4 against.